

Doctor Appointment Booking System

Project Report

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Project Team

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Year: IV B.Tech

Database: MongoDB

Frontend: React, Tailwind CSS

Backend: Node.js, Express.js

Abstract

MediCare is a MERN-stack based smart doctor appointment booking system that enables patients to search doctors by specialization, register securely, book appointments, and track appointment history. Doctors can manage bookings, confirm or decline requests, and update their professional profiles. The system provides a responsive interface, secure JWT authentication, and MongoDB-based data management.

Technologies Used

- React.js
- Vite
- Tailwind CSS
- Node.js
- Express.js
- MongoDB
- JWT
- Axios
- bcrypt
- REST API

Main Features

- Patient Registration/Login
- Doctor Registration/Login
- Doctor Search by Specialization
- Appointment Booking
- Appointment History
- Doctor Dashboard
- Booking Confirmation/Decline
- Responsive UI
- JWT Authentication
- MongoDB Database

Modules

Patient Module:

- Register/Login
- Search Doctors
- Book Appointment
- View & Cancel Appointments

Doctor Module:

- Login
- Manage Profile
- Confirm/Decline Bookings
- View Upcoming Consultations

Database

MongoDB collections include Users, Doctors, Appointments and Authentication data.

Sample Screenshots

Description:

The system dashboard displays the available doctors, appointment booking options, patient registration and login modules, doctor dashboard, and appointment management features. Users can browse doctors, schedule consultations, and monitor appointment status in real time.

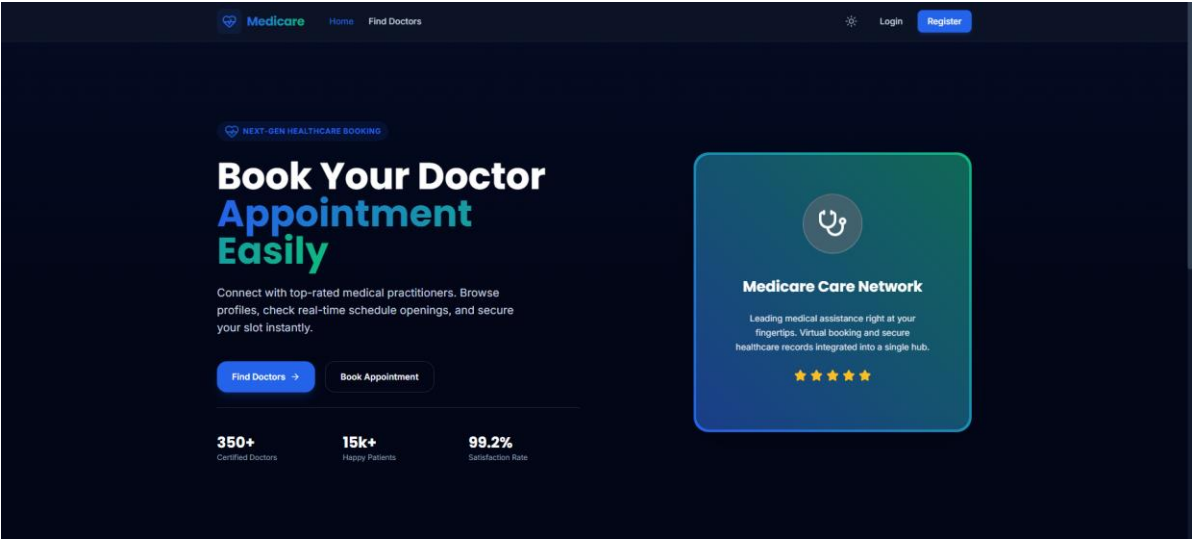


Image 2 – Speciality Filtering and Service Features

Description: The system dynamically categorizes and filters healthcare practitioners based on medical specialties selected by the user. The UI displays interactive specialty cards alongside core platform capabilities—ensuring verified licensing, immediate slot booking synchronization, and transparent patient management details.

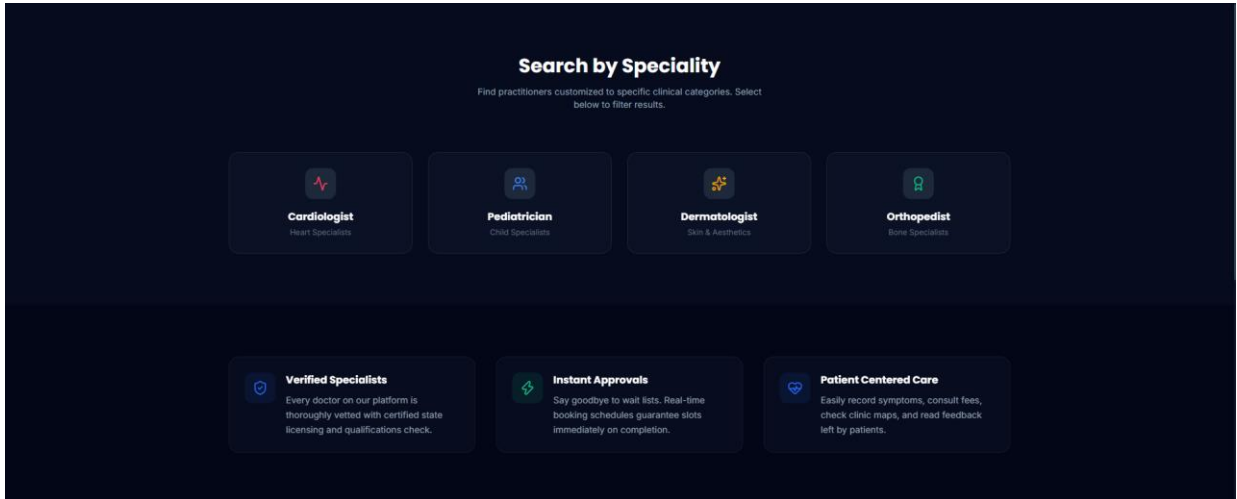


Image 3 – Top Certified Doctors Directory

Description: Displays top-rated healthcare experts with interactive profile cards. Each card highlights key details including real-time availability badges, star ratings with patient review counts, medical specialties, educational qualifications, years of experience, affiliated hospital locations, and consultation fees, with direct options to view full practitioner profiles.

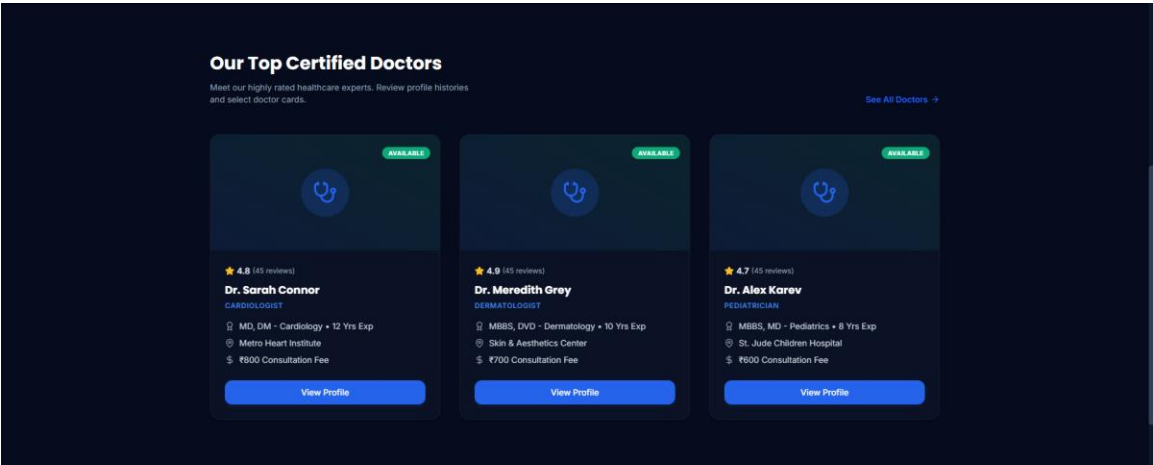


Image 4 – Authentication Module & Role Simulation

Description: Presents the application's authentication portal equipped with JWT-based session management and secure password encryption. The UI integrates one-click demo account auto-fill functionality for Patient, Doctor, and Admin roles, facilitating seamless access control testing and role-based interface navigation.

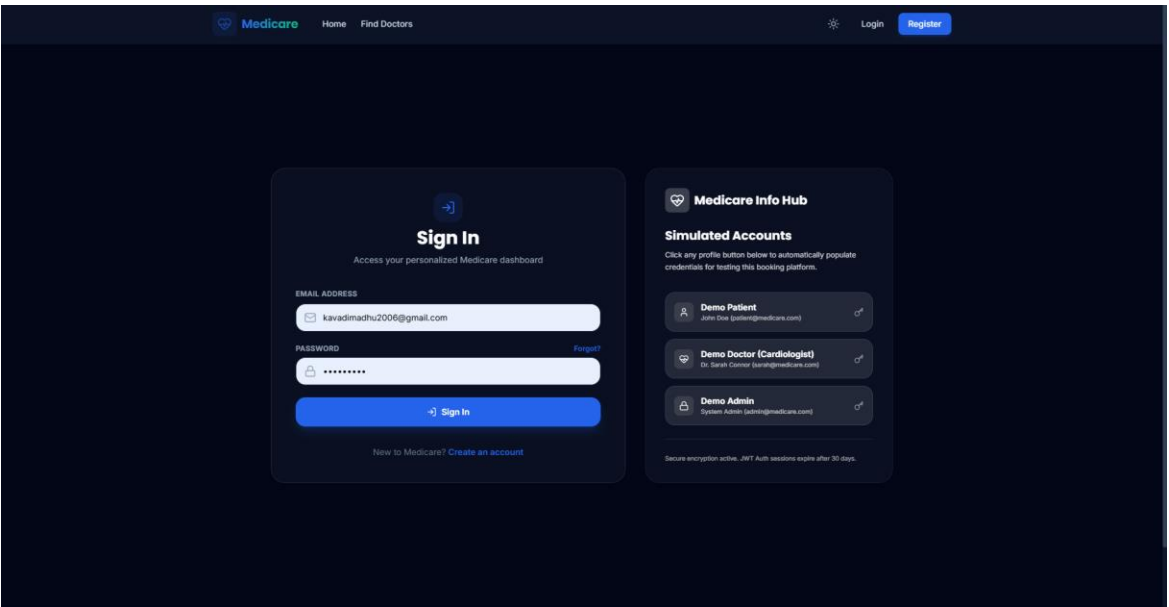
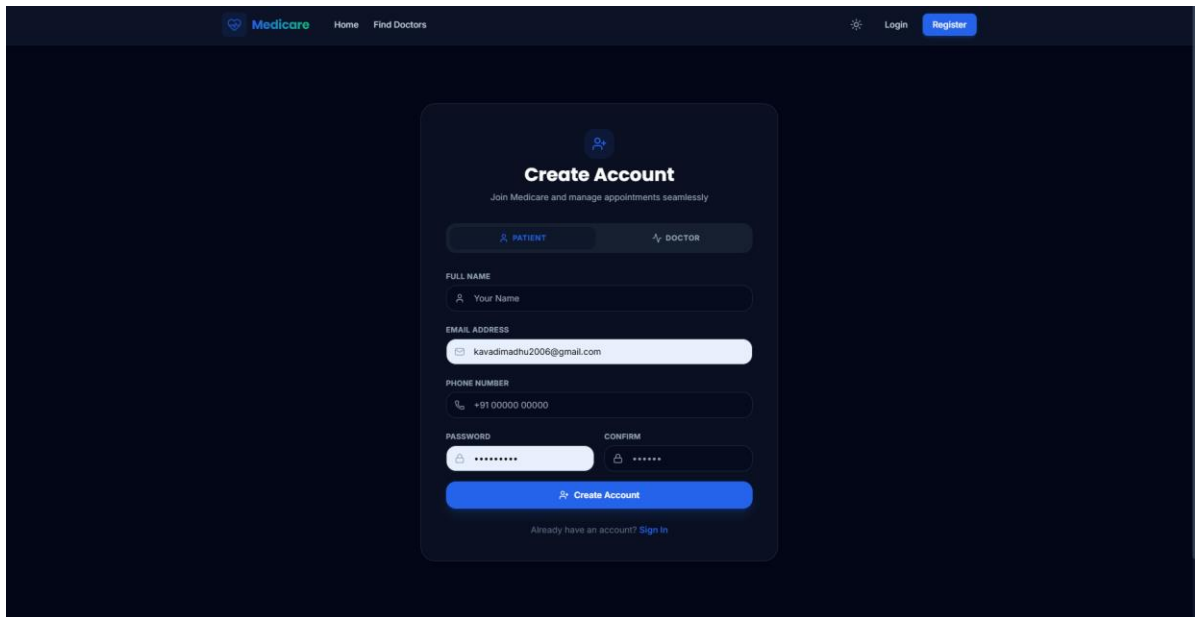


Image 5 – Account Registration Portal

Description: The registration page enables new users to join the platform and manage healthcare appointments. The interface includes a role selection toggle allowing users to register as either a Patient or a Doctor, alongside input fields for full name, email address, phone number (+91 Indian standard format), password, and password confirmation.



The screenshot displays the 'Create Account' page of the Medicare platform. The page has a dark blue background. At the top, a navigation bar contains the Medicare logo, 'Home', 'Find Doctors', a sun icon, 'Login', and a 'Register' button. The main content area features a central white card with the title 'Create Account' and the subtitle 'Join Medicare and manage appointments seamlessly'. Below this, there is a toggle switch for 'PATIENT' (selected) and 'DOCTOR'. The form includes input fields for 'FULL NAME' (placeholder: 'Your Name'), 'EMAIL ADDRESS' (placeholder: 'kavadinadhu2006@gmail.com'), 'PHONE NUMBER' (placeholder: '+91 00000 00000'), 'PASSWORD' (placeholder: '*****'), and 'CONFIRM' (placeholder: '*****'). A blue 'Create Account' button is at the bottom of the form, with a link 'Already have an account? Sign In' below it.

Image 6 – My Appointments Management Dashboard

Description: The patient's appointment management portal allows users to track consultation history, check active schedules, or cancel slots. The interface features tabbed navigation filters (ALL, UPCOMING, COMPLETED, CANCELLED), detailed booking cards displaying doctor profiles, appointment dates, times, reported symptoms (e.g., *High Fever, Acne problem*), real-time status tags (PENDING), and action buttons for booking cancellation or scheduling a new doctor.

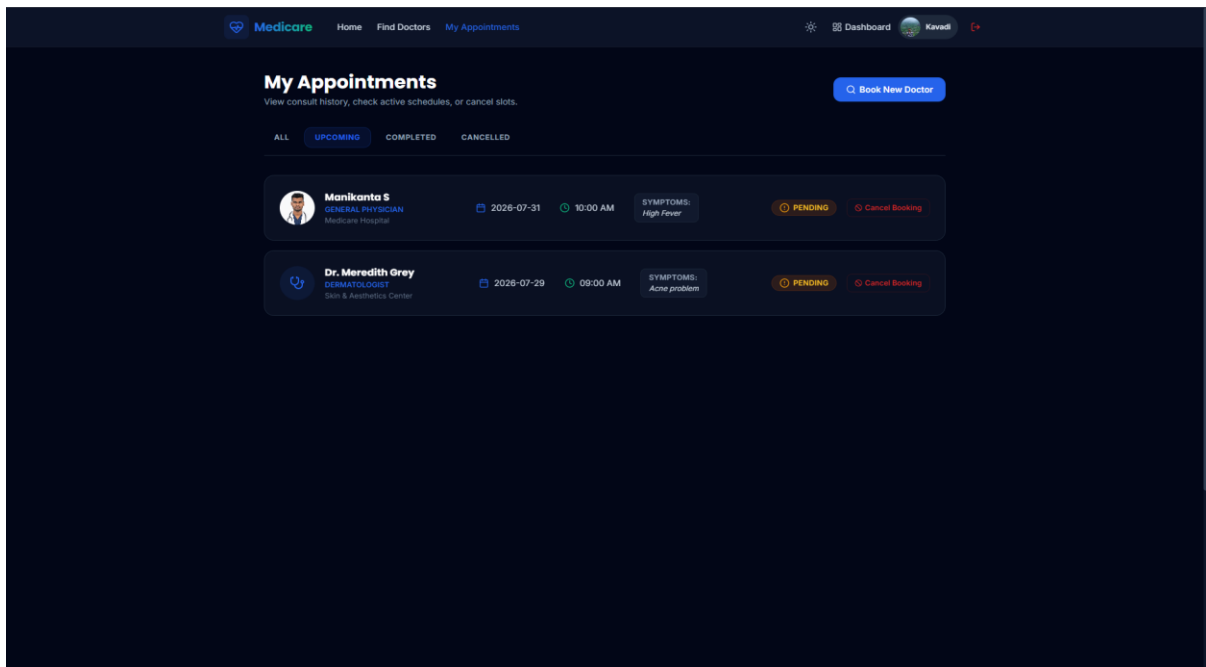
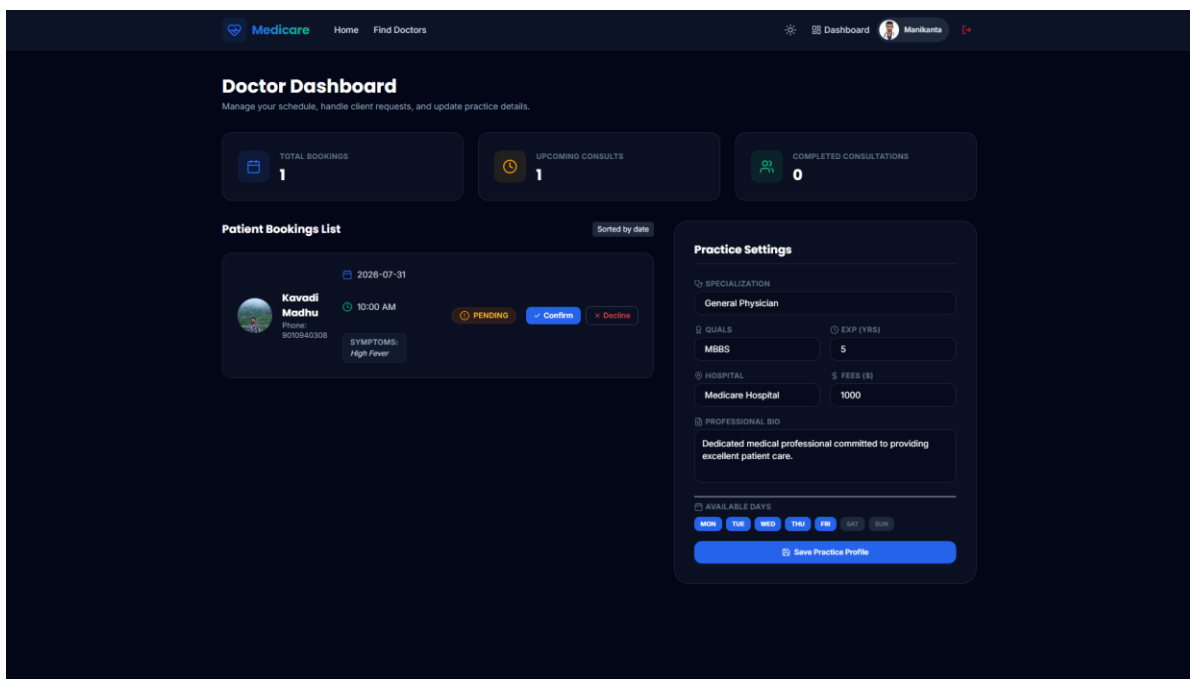


Image 7 – Doctor Portal & Practice Management Dashboard

Description: The Doctor Dashboard allows medical practitioners to manage their daily schedules, handle incoming patient booking requests, and configure practice details. The page features real-time metric cards (Total Bookings, Upcoming Consults, Completed Consultations), a **Patient Bookings List** with action controls to Confirm or Decline pending appointments, and a **Practice Settings** panel to update specialization, qualifications, consultation fees, professional bio, and available practice days.



USED THESE COMMANDS TO RUN :

Step 1: Run the Backend

```
cd backend
```

```
npm install
```

```
npm run dev
```

Step 2: Run the Frontend

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Your browser will open the frontend at <http://localhost:5173>.

Step 1: Run the Backend

In your VS Code terminal, run:

```
powershell  
  
cd backend  
npm install  
npm run dev
```

Step 2: Seed the Database (Optional, to populate sample listings/users)

While the backend terminal is running, open a **new terminal tab** (click the  icon in the terminal window header) and run:

```
powershell  
  
cd backend  
npm run seed
```

Step 3: Run the Frontend

In that same second terminal window (or a new one), navigate to the frontend folder and start the development server:

```
powershell  
  
cd ../frontend  
npm install  
npm run dev
```

Your browser will open the frontend at <http://localhost:5173>.

